

PF Lab 9 Tasks

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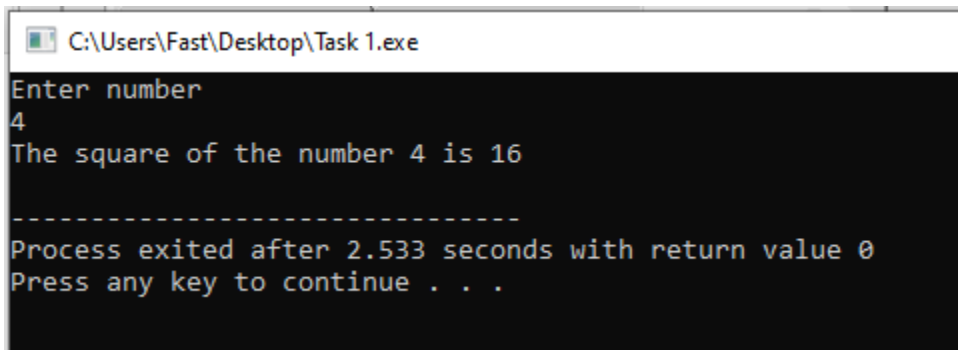
Section: BCS-1K

Task 1

```
#include<stdio.h>
```

```
int square(int num){  
    int s;  
    s=num*num;  
    return s;  
}
```

```
int main(){  
    int num;  
    printf("Enter number\n");  
    scanf("%d",&num);  
    printf("The square of the number %d is %d\n",num,square(num));  
    return 0;  
}
```



```
C:\Users\Fast\Desktop\Task 1.exe  
Enter number  
4  
The square of the number 4 is 16  
-----  
Process exited after 2.533 seconds with return value 0  
Press any key to continue . . .
```

Task 2

```
#include<stdio.h>
```

```
int main()
```

```
{
```

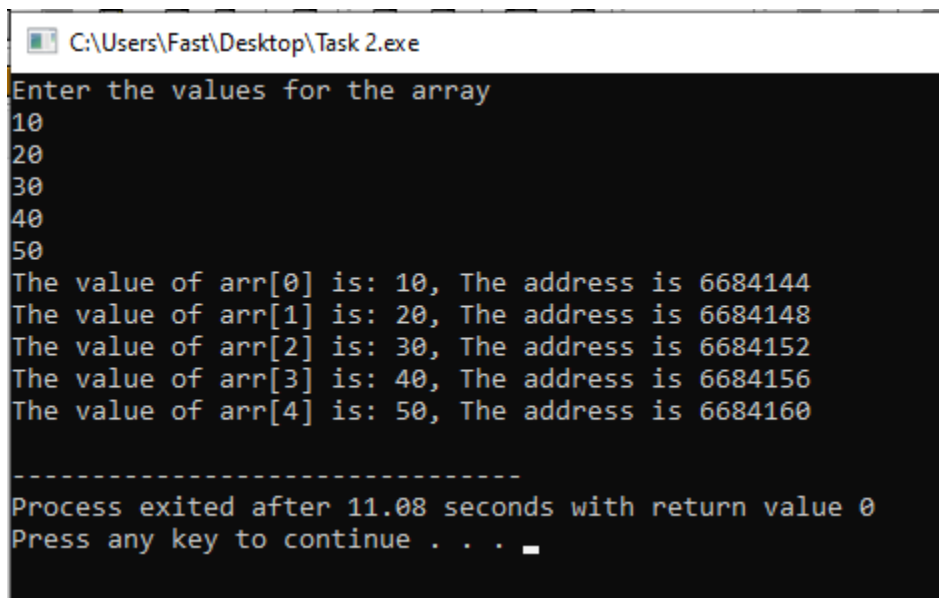
```
    int arr[5];
```

```
    printf("Enter the values for the array\n");
```

```

for(int i=0;i<5;i++){
    scanf("%d",&arr[i]);
}
int *p=arr;
for(int i=0;i<5;i++){
    printf("The value of arr[%d] is: %d, The address is %d\n",i,*(p+i), p+i);
}
return 0;
}

```



```

C:\Users\Fast\Desktop\Task 2.exe
Enter the values for the array
10
20
30
40
50
The value of arr[0] is: 10, The address is 6684144
The value of arr[1] is: 20, The address is 6684148
The value of arr[2] is: 30, The address is 6684152
The value of arr[3] is: 40, The address is 6684156
The value of arr[4] is: 50, The address is 6684160

-----
Process exited after 11.08 seconds with return value 0
Press any key to continue . . .

```

Task 3

```
#include<stdio.h>
```

```

void displayMatrix(int arr[2][3]){
    printf("The matrix is:\n");
    for(int i=0;i<2;i++){
        for(int j=0;j<3;j++){
            printf("%d ",arr[i][j]);

```

```
        }  
        printf("\n");  
    }  
}
```

```
int calculateSum(int arr[2][3]){  
    int sum=0;  
    for(int i=0;i<2;i++){  
        for(int j=0;j<3;j++){  
            sum+=arr[i][j];  
        }  
    }  
    printf("The sum is %d\n",sum);  
    return 0;  
}
```

```
float calculateAverage(int arr[2][3]){  
    int total=0;  
    float avg;  
    for(int i=0;i<2;i++){  
        for(int j=0;j<3;j++){  
            total+=arr[i][j];  
        }  
    }  
    avg=total/6;  
    printf("The average is %.2f\n",avg);  
    return 0;  
}
```

```
int findMax(int arr[2][3]){  
    int max=0;  
    for(int i=0;i<2;i++){  
        for(int j=0;j<3;j++){  
            if(arr[i][j]>max){  
                max=arr[i][j];  
            }  
        }  
    }  
    printf("The largest element is %d\n",max);  
    return 0;  
}
```

```
int main()  
{  
    int arr[2][3]={{1,2,3},{4,5,6}};  
    displayMatrix(arr);  
    calculateSum(arr);  
    calculateAverage(arr);  
    findMax(arr);  
    return 0;  
}
```

```
C:\Users\Fast\Desktop\Task 3.exe
The matrix is:
1 2 3
4 5 6
The sum is 21
The average is 3.00
The largest element is 6

-----
Process exited after 0.03014 seconds with return value 0
Press any key to continue . . .
```

Task 4

```
#include<stdio.h>
```

```
int findMax(int arr[],int size){
    int max=0;
    for(int i=0;i<size;i++){
        if(arr[i]>max){
            max=arr[i];
        }
    }
    printf("The largest element is %d\n",max);
    return 0;
}
```

```
int main()
{
    int size=5;
    int arr[size]={2,4,5,7,8};
    findMax(arr,size);
    return 0;
}
```

 C:\Users\Fast\Desktop\Task 4.exe

The largest element is 8

Process exited after 0.03343 seconds with return value 0

Press any key to continue . . . 